1.Studying gases helped the development of chemistry, how ?

The Earth consist of gases, and it concentrate by chilling. Gas state is basic and more simple state. Gases create pressure and heat. It can spread somewhere easily. In Chemistry, People use gases for increasing temperature , increasing or decreasing pressure, creat new reactions, etc. (Gases also allow measurement.)

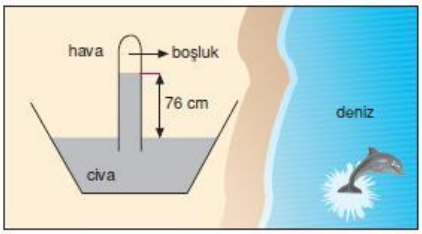
2. What was the basis of all things up until the 18th century?

Earth, water, air and fire was the basis of all things up until 18th century. There wasn't scientific basis until 18th century and not a theory for chemistry.

3. Name the first few elements identified ?

People approve 4 elements when the first centuries. Their names are earth, air, water and fire but they also know that Earth has many minerals.
They recognized elements first in 18th century. The first element is Arsenic (As), second is Antimony (Sb), Bismuth(Bi), Phosphorus(P), Cobalt(Co), Platinum(Pt), Nickel(Ni), Nitrogen(N), Chlorine(Cl), and so on.

4. How does day to day differences change the height of the Mercury in the barometer?



If the height and temperature are changed, the height of mercury will change. It means the balance of pressure for gases.

$P = h.d.g$ $\quad d_{civa} = 13.6\ g/cm^3$

$P = 76.13.6.1$

$P = P_0 = 1033{,}6\ g.f/cm^2$

5. What is the principle of Thermometery explain by drawing ?

A thermometer is a devide that measures the temperature of things. The liquid in the middle of the thermometer moves up or down depending on the temperature.

6. What is the conservation of mass in a reaction, why is it significant in the development of chemistry ?

In a chemical reaction, matter is neither created nor destroyed. According to the law of conservation of mass, the mass of the products in a chemical reaction must equal the mass of the reactants. It's useful concept in chemistry because scientists can compare the system .

7.The volume of a gas IS INVERSELY proportional to the pressure exerted upon it. Explain ?

Pressure x Volume = Constant
If the volume decreases, the molecules encounter the container walls more often. If the volme increases, the rate of collisions and the pressure decreases.

8. Why do you think the development of analytical balance is significant ?

People use the analytical balance in chemistry and medicine. It's important because it is used for making medicine, and human health sometimes depends on medicine.

9. What is called "Phlogisticated Air" and "Dephlogisticated Air" and why ?

The mixture of nitrogen and carbon dioxide was called Phlogisticated Air. On the other hand, when oxygen was first discovered, it was called Dephlogisticated Air because it had a large affinity for phlogiston, it must be particularly devoid of it.

10. DEPHLOGISTICATED Air supports the life and the PHLOGISTICATED Air does not.

11. Why do we need a spark of electricity for the combustion of Hydrogen and Oxgyen?

Hydrogen and oxygen react in a chemical formula to synthesise water. The spark is the activation energy for begining his reaction. Breaking bonds takes in energy. Creating bonds releases energy.

$$2H_2 + O_2 \longrightarrow 2H_2O$$

12. What is the principle of Eudiometer ?

Aplications of a eudiometer include the analysis of gases and determination of volume differences in chemical reactions.
The eudiometer is filled with water, inverted so that its open end is facing the ground and then submersed in a basic of water.
A chemical reaction is taking place through which gas is created.

13. How did Lavoisier discovered H and O ?

Lavoisier observed that there were quantitative reactions involving heat. Heat added to water gave steam discovered that air reacting to make calx lost 1/6 its volume. He thought air contains an element (Oxygen). Lavoisier named Cavendish's phlogiston hydrogen. Lavoisier concluded that the reaction of hydrogen with air to make dew was actually the combining of hydrogen and oxygen to make water.

14. Why is nomenclature and naming system is important in Science, Use your own words.

It makes the science more understandable. For example, If someone wants to get information about computer science, he/she will also learn some words about computers.

15. Explain Dalton's particle chemistry ?

- All matter is made of atoms. Atoms are indivisible and indestructible.
- All atoms of a given element are identical in mass and properties.
- Compounds are formed by a combination of two or more different kinds of atoms.
- A chemical reaction is a rearrangement of atoms.

16. How did atoms combine to make different compounds ?

Most of atoms want to become stable, they must bond with other atoms (that are available). Stable atoms have complete outer orbitals. Since metallic atoms typically have excess electrons in their outer orbitals, they tend to try and donate electrons to nonmetals that called ionic bonding. Nonmetal elements are typically short electrons and will covalently bond to each other by sharing electrons.

17. Explain Law of Multiple Proportions and how is it related to Law of Partial Pressures ?

The law of multiple proportions says that when two elements combine with each other to form two or more compounds, the ratios of the masses of one element that combines with the fixed mass of the other are simple whole numbers.
Law of Partial Pressures says that the total pressure exerted by a mixture of gases is equal to the sum of the partial pressures of the gases in the mixture. -- $PV \alpha n$ --

18. At what AGE do you think we are living now and why ?

I think we are living in technological age because there are many technological items which are developed everyday. And also people don't have to memorize many different information they can find them on the Internet.

19. Rapid development of chemical industry is fostered by THE COMMERCIAL SUCCESS OF CHEMISTRY
Draw a parallel for today in your field.

20. What do you expect for the Future of Science and Technology? User your own words .

People may find a treatment for cancer and other diseases, maybe we can avoid to be hungry eating pill. As everybody wants, robots are developed and they can do what we need.

21. Explain Electrolysis ?

Electrolysis is a process which break down ionic materials using electrical current.

Here is what happens when during electrolysis:

- Positively charged ions move to the negative electrode during electrolysis. They receive electrons and are reduced.
- Negatively charged ions move to the positive electrode during electrolysis. They lose electrons and oxidised.

22. How can we determine the composition starts ?

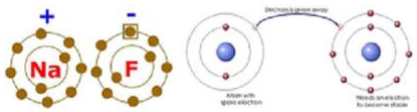
They are determined through spectroscopy. Spectroscopy is the study of something using spectre.

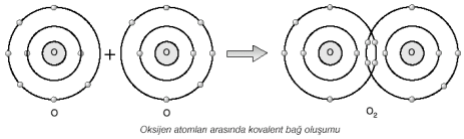
23. What is chemical algebra, describe with an example ?

Chemical algebra is also known as Stoichiometry. It is founded on the law of conservation of mass where the total mass of the reactants equals the total mass of products.

$$CH4 + 2\ O2 \rightarrow CO2 + 2\ H2O.$$

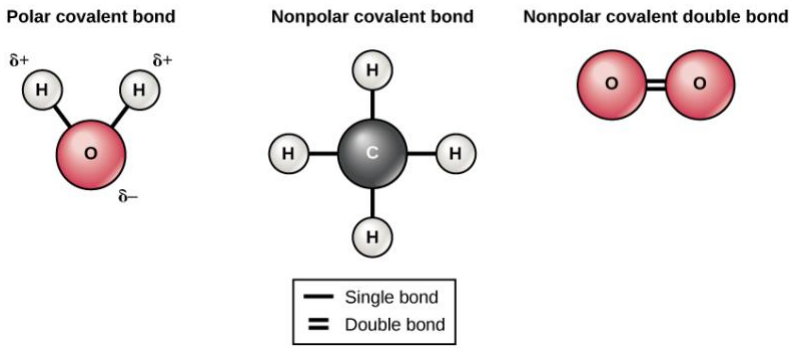
24. Explain the 2 types of bonding through electrons ?





Ionic bond is formed when a non metal and metal exchange electrons . Covalent bond is formed when electrons are shared between two non metals.

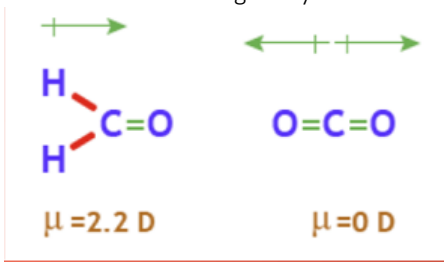
25. What is the polar and non-polar bonding ?



If electrons shared equally, the bond is nonpolar. If electrons shared unequally, the bond is polar.

26. What is the dipole moment and the implications ?

A large difference in electronegativity between two bonded atoms will cause a dipole.



27. How can a spectroscope be used in characterization of elements ?

Various elements were known to give discreate spectra when heated.
By dipping a platinum wire into a solution containing an element and then placing it in a flame.

28. Explain periodic table and significance ?

The periodic table organizes elements according to similar properties so you can tell the characteristics of an element just by lookig at its position on the table.

29. What are the common elements of organic matters ?

Carbon (C), and Hydrogen(H), and Oxygen(O).

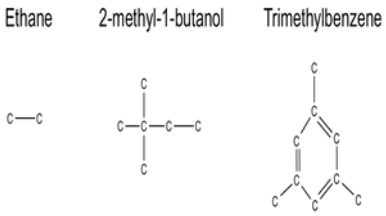
30. Different chemical compounds but same composition of elements. How do you explain ?

When the elements are joined, the atoms lose their individual properties and have different properties from the elements they are composed of. A chemical formula is used a quick way to show the composition of compounds. Letters, numbers, and symbols are used to represent elements and the number of elements in each compound.

31. One of the areas where computer scientist work is the prediction of 3-D structure of molecules. What do they consider in these estimations.

32. What is carbon skeleton. Give a few examples for it.

Carbon could have 4 bonds. It forms a chain by connecting to itself or to other atoms to make organic molecules.



33. How did the discover of Robert Hooke help the development of biology ?

Cells are discovered by Robert Hooke and he investigate cells in 17th century. He also invented the microscope.

34. Name few advances in the microscopy technology.

- Poor quality glass
- Lens acting as prism distorts light(chromatic effect)
- The electron microscope advanced biological research
- 20 times as sensitive as the light microscope

35. Briefly explain cell theory with respect to plant and animal cells.

The Cell Theory is

- All living organisms are composed of cells. They may be unicellular or multicellular.
- The cell is the basic unit of life.
- Cells arise from pre-existing cells.

Larger organisms, such as a dog or a tomato plant, are also made up of cells. You could take a small tissue sample and place it under a microscope to see one actual cell within the larger living thing.

36. What is miasma theory ?

The miasma theory states that diseases and sicknesses come from breathing in a miasma. It would effect individuals within the locale that gave rise to such vapors.

37. Explain the work of Louis Pasteur and its significance.

Louis Pasteur's pasteurization experiment illustrates the fact that the spoilage of liquid was caused by particles in the air rather than the air itself. These experiments were important pieces of evidence supporting the idea of Germ Theory of Disease.

38. What is struggle for survival ?

Darwin noted on the struggle for survival.
Struggle for existence inevitably increase variety.
Every being musr suffer destruction during some period of its life. Otherwise, on the principal of geometrical increase, no land could support the product.

39. What is the principles of natural selection ?

- Populations grow exponentially
- Variation occurs within species
- Traits can be inherited

40. Explain Mendels work and comment on his due dillegence.

Mendel work about genetics.
He grow and studied 29.000 pea plants.
For his experiments, he selected the common pea plant and in preliminary testing identified 7 pairs of characteristic for study (form, color, length, etc.)

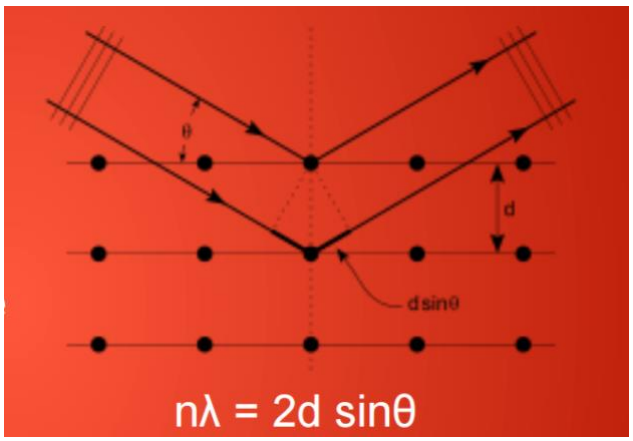
41. Why do you think Morgon used Drosophila in his studies.

- Fruit flies have a shotr life cycle that lasts an average of just 30 days.
- Fruit flies produce huge numbers of offspring.
- Consideration is cost.

42. What are the landmark achievements that led the development of Molecular Biology?

- The existence of DNA
- The elucidation of the structure of DNA
- The understanding how DNA controls the structure of the proteins made by cells.

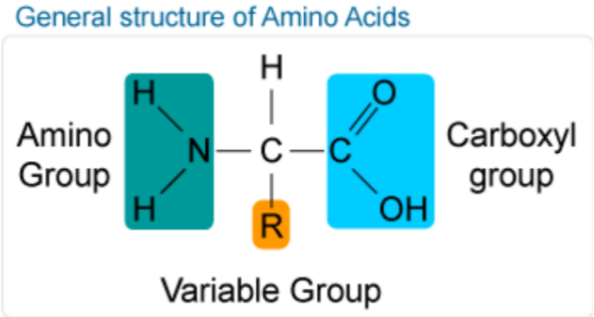
43. Explain X-Ray diffraction and its significance



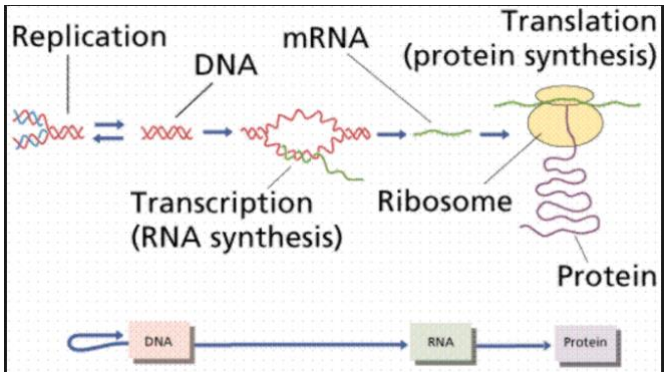
Bragg's studied the behavior of X-Rays when shined on crystals. Bragg determined a mathematical relationship between the wavelength of the X-Rays and the distance between the atoms in a crystal through reflection.

44. What is the basic structure of Amino Acids

All amino acids have the same basic structure, an acid group at one end and an amine group at the other.



45. Mapping from DNA to protein. Briefly explain



46. Briefly explain Protein Structures.

Primary – Exact sequence of amino acids before folding
Secondary – Simple folding create simple structures
Tertiary – Folding results in complex 3D structures
Quaternary – Multiple 3D subunits orginazed into a bigger structure

47. Explain the relationship between biochemistry, molecular biology and genetics

- Biochemistry focused on the protein part of life functions. It studies the components independent of the organism.
- Genetics focuses on the gene part. Usually mutants are used. So, it is organism without the component.
- Molecular Biology integrates those two, as can be quite well ascertained from the “central dogma”.

48. Comment on the progress of number theory. Feel free to read on the web

Number theory is about integers. Computers compute big integers that means approximately infinite. Nowadays it is used for communication systems, network design, high speed computers, coding system etc.

49. Briefly explain the Early Computer of Babbage

The Analytical Engine incorporated an arithmetic logic unit, control flow in the form of conditional branching and loops, and integrated memory, making it the first design for a general purpose computer.
The input, consisting of programs and data was to be provided to the machine via punchd cards, a method being used at the time to direct mechanical looms such as Jacquard loom.

50. Explain AND as well as OR circuit

AND OPERATOR	OR OPERATOR
1-1 = 1	1-1 = 1
1 -0 = 0	1-0 =1
0-1 = 0	0-1=1
0-0=0	0-0 =0

51. Briefly explain Turing Machine with an example.

Alan Turing published a landmark paper on computing that defined a machine that would be capable of solving any conceivable mathematical problem for which an algorithm could be written.

Example

Turing machine $M = (Q, \Gamma, \Sigma, \delta, s, B, F)$ with

- $Q = \{q_0, q_1, q_2, q_3, q_4\}$,
- $\Gamma = \{a, b, X, Y, \#\}$,
- $\Sigma = \{a, b\}$,
- $s = q_0$,
- $B = \#$,
- δ given by

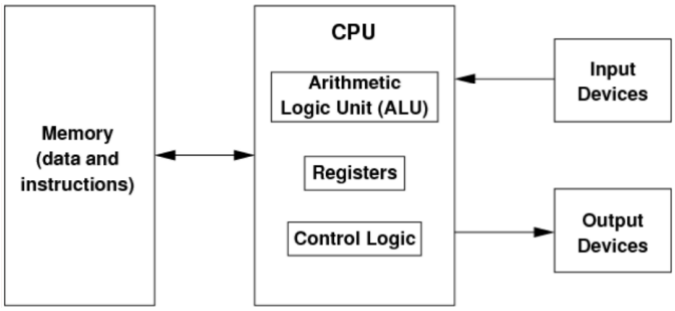
	a	b	X	Y	$\#$
q_0	(q_1, X, R)	–	–	(q_3, Y, R)	–
q_1	(q_1, a, R)	(q_2, Y, L)	–	(q_1, Y, R)	–
q_2	(q_2, a, L)	–	(q_0, X, R)	(q_2, Y, L)	–
q_3	–	–	–	(q_3, Y, R)	$(q_4, \#, R)$
q_4	–	–	–	–	–

52. Why is Turing Machine in the development of computer programming

English effort in cryptography and helped break the German Enigma Code. Turing built an electromechanical machine that speeded up the breaking of coded message. Breaking the German code gave the British and Americans the ability to know where the German submarines are stationed.

53. What is Von Neumann Architecture ?

The Von Neumann architecture is the basic design for all modern computers.



54. What is Incompleteness Theorem ?

Kurt Gödel eliminated the possibility that any system of mathematics could be complete. Gödel proved that in any system there would be theorems that were true but could never be proven and theorems that were false but could never be disproven.

55. What is an NP-Complete problem?

Any given solution to an NP-complete problem can be verified quickly (in polynomial time), the most notable characteristic of NP-complete problems is that no fast solution to them is known.